09.5_modelos_hiperparametros_tecnica

May 3, 2026

[1]:
```python
from pathlib import Path
import pandas as pd
import numpy as np
```

[2]:
```python
PROJECT_ROOT = Path("/home/harielpadillasanchez/Documentos/TT/TT2")

FEW_DIR = PROJECT_ROOT / "outputs" / "comparison_finalists"
ZERO_DIR = PROJECT_ROOT / "outputs" / "comparison_finalists_zero"
OUT_DIR = PROJECT_ROOT / "outputs" / "comparison_prompting_final"
OUT_DIR.mkdir(parents=True, exist_ok=True)

FEW_FILE = FEW_DIR / "ranking_global.csv"
ZERO_FILE = ZERO_DIR / "ranking_global_zero.csv"

print("FEW_FILE :", FEW_FILE)
print("ZERO_FILE:", ZERO_FILE)
print("OUT_DIR  :", OUT_DIR)
print("Existe FEW_FILE :", FEW_FILE.exists())
print("Existe ZERO_FILE:", ZERO_FILE.exists())
```

```
FEW_FILE : /home/harielpadillasanchez/Documentos/TT/TT2/outputs/comparison_final
ists/ranking_global.csv
ZERO_FILE: /home/harielpadillasanchez/Documentos/TT/TT2/outputs/comparison_final
ists_zero/ranking_global_zero.csv
OUT_DIR  :
/home/harielpadillasanchez/Documentos/TT/TT2/outputs/comparison_prompting_final
Existe FEW_FILE : True
Existe ZERO_FILE: True
```

[3]:
```python
df_few = pd.read_csv(FEW_FILE)
df_zero = pd.read_csv(ZERO_FILE)

print("Shape few :", df_few.shape)
print("Shape zero:", df_zero.shape)

display(df_few.head(3))
display(df_zero.head(3))
```

```
Shape few : (46, 28)
Shape zero: (50, 30)

  model_key  config_label ruleset  gen_temperature  gen_top_p  \
0    llama3  llama3_cfg_3      R1              0.3        0.9
1    llama3  llama3_cfg_2      R2              0.7        0.9
2    llama3  llama3_cfg_2      R0              0.2        0.9

   gen_repetition_penalty  gen_max_new_tokens       sari       bleu  \
0                    1.15               256.0  42.880052  14.297969
1                    1.10               512.0  42.969879  15.394757
2                    1.10               256.0  43.080279  15.378169

   fernandez_huerta_pred  …  rouge1_f  rouge2_f  rougeL_f  inflesz_pred  \
0              82.165161  …  0.477212  0.284198  0.417246     64.355483
1              84.192944  …  0.479850  0.283303  0.398782     66.226652
2              81.424226  …  0.489988  0.299905  0.411665     66.517505

   inflesz_src  inflesz_delta  bertscore_f1   owner  leader_score  \
0    51.983615      12.371868      0.807136  hariel     58.013464
1    51.983615      14.243037      0.805323   nancy     57.994847
2    51.983615      14.533891      0.803035  hariel     57.969580

                                  config_signature
0  hariel | llama3 | llama3_cfg_3 | R1 | temp=0.3…
1  nancy | llama3 | llama3_cfg_2 | R2 | temp=0.7 …
2  hariel | llama3 | llama3_cfg_2 | R0 | temp=0.2…

[3 rows x 28 columns]

    owner model_key           config_label ruleset  gen_temperature  gen_top_p  \
0  hariel    llama3   hariel_llama3_cfg_1      R0              0.2       0.85
1   nancy   mistral   nancy_mistral_cfg_2      R4              0.3       0.90
2  hariel   mistral  hariel_mistral_cfg_1      R4              0.3       0.90

   gen_repetition_penalty  gen_max_new_tokens gen_do_sample  \
0                    1.05                 256           NaN
1                    1.15                 400          True
2                    1.10                 256           NaN

   gen_no_repeat_ngram_size  …  deletions_proportion  rouge1_f  rouge2_f  \
0                       NaN  …              0.509976  0.483206  0.288552
1                       4.0  …              0.462206  0.472995  0.280721
2                       NaN  …              0.443968  0.457271  0.257443

   rougeL_f  inflesz_pred  inflesz_src  inflesz_delta  bertscore_f1  \
0  0.405020     73.239084    51.983615      21.255469      0.802757
1  0.396466     62.653191    51.983615      10.669577      0.801483
2  0.383889     58.801009    51.983615       6.817394      0.799005
```

```
   leader_score                                    config_signature
0     57.300870  hariel | llama3 | hariel_llama3_cfg_1 | R0 | t…
1     57.201885  nancy | mistral | nancy_mistral_cfg_2 | R4 | t…
2     56.748324  hariel | mistral | hariel_mistral_cfg_1 | R4 |…

[3 rows x 30 columns]
```

[4]:
```python
df_few["prompt_family"] = "few-shot"
df_zero["prompt_family"] = "zero-shot"

print(df_few["prompt_family"].unique().tolist())
print(df_zero["prompt_family"].unique().tolist())
```

```
['few-shot']
['zero-shot']
```

[5]:
```python
all_cols = sorted(set(df_few.columns).union(set(df_zero.columns)))

for col in all_cols:
    if col not in df_few.columns:
        df_few[col] = np.nan
    if col not in df_zero.columns:
        df_zero[col] = np.nan

df_prompt = pd.concat([df_few[all_cols], df_zero[all_cols]], ignore_index=True,␣
 ↪sort=False)

print("Shape combinado prompting:", df_prompt.shape)
display(df_prompt.head(5))
```

```
Shape combinado prompting: (96, 31)

   additions_proportion  bertscore_f1       bleu  compression_ratio_eval  \
0              0.391925      0.807136  14.297969                0.861692
1              0.352626      0.805323  15.394757                0.781710
2              0.325193      0.803035  15.378169                0.807718
3              0.338931      0.800380  14.955279                0.761530
4              0.264534      0.807977  17.534299                0.761265

   config_label                                    config_signature  \
0  llama3_cfg_3  hariel | llama3 | llama3_cfg_3 | R1 | temp=0.3…
1  llama3_cfg_2  nancy | llama3 | llama3_cfg_2 | R2 | temp=0.7 …
2  llama3_cfg_2  hariel | llama3 | llama3_cfg_2 | R0 | temp=0.2…
3  llama3_cfg_2  nancy | llama3 | llama3_cfg_2 | R1 | temp=0.7 …
4  llama3_cfg_1  hariel | llama3 | llama3_cfg_1 | R2 | temp=0.2…

   deletions_proportion  exact_copy  fernandez_huerta_delta  \
0              0.481467         0.0               -6.149181
```

```
1              0.502511        0.0               -4.121398
2              0.469978        0.0               -6.890116
3              0.500175        0.0               -4.549150
4              0.443409        0.0               -2.313630

   fernandez_huerta_pred  …  levenshtein_similarity model_key   owner  \
0              82.165161  …                0.465548    llama3  hariel
1              84.192944  …                0.490718    llama3   nancy
2              81.424226  …                0.523537    llama3  hariel
3              83.765192  …                0.481710    llama3   nancy
4              86.000712  …                0.543717    llama3  hariel

   prompt_family  rouge1_f  rouge2_f  rougeL_f  ruleset       sari  \
0       few-shot  0.477212  0.284198  0.417246       R1  42.880052
1       few-shot  0.479850  0.283303  0.398782       R2  42.969879
2       few-shot  0.489988  0.299905  0.411665       R0  43.080279
3       few-shot  0.467946  0.286316  0.418976       R1  42.769451
4       few-shot  0.497984  0.318418  0.443558       R2  42.218842

   sentence_splits
0             0.70
1             0.30
2             0.70
3             0.55
4             0.65

[5 rows x 31 columns]
```

[6]:

```python
optional_cols = [
    "gen_temperature",
    "gen_top_p",
    "gen_repetition_penalty",
    "gen_max_new_tokens",
    "gen_do_sample",
    "gen_no_repeat_ngram_size",
]

for col in optional_cols:
    if col not in df_prompt.columns:
        df_prompt[col] = np.nan

df_prompt["gen_do_sample"] = df_prompt["gen_do_sample"].astype("string").
    ↪fillna("NA")
df_prompt["gen_no_repeat_ngram_size"] = df_prompt["gen_no_repeat_ngram_size"].
    ↪astype("string").fillna("NA")

print("Columnas opcionales normalizadas.")
```

Columnas opcionales normalizadas.

[7]:
```python
df_prompt["leader_score"] = (
    0.6 * df_prompt["sari"] +
    0.4 * (df_prompt["bertscore_f1"] * 100.0)
)

display(
    df_prompt[
        [c for c in [
            "prompt_family",
            "owner",
            "model_key",
            "config_label",
            "ruleset",
            "sari",
            "bertscore_f1",
            "leader_score",
            "compression_ratio_eval",
            "exact_copy",
        ] if c in df_prompt.columns]
    ].head(10)
)
```

```
  prompt_family   owner model_key   config_label ruleset       sari  \
0      few-shot  hariel    llama3  llama3_cfg_3      R1  42.880052
1      few-shot   nancy    llama3  llama3_cfg_2      R2  42.969879
2      few-shot  hariel    llama3  llama3_cfg_2      R0  43.080279
3      few-shot   nancy    llama3  llama3_cfg_2      R1  42.769451
4      few-shot  hariel    llama3  llama3_cfg_1      R2  42.218842
5      few-shot  hariel    llama3  llama3_cfg_1      R4  41.435601
6      few-shot   nancy    llama3  llama3_cfg_1      R0  41.540371
7      few-shot   nancy    llama3  llama3_cfg_1      R4  42.008218
8      few-shot  hariel    llama3  llama3_cfg_3      R0  41.658159
9      few-shot  hariel    llama3  llama3_cfg_2      R1  41.371788

   bertscore_f1  leader_score  compression_ratio_eval  exact_copy
0      0.807136     58.013464                0.861692         0.0
1      0.805323     57.994847                0.781710         0.0
2      0.803035     57.969580                0.807718         0.0
3      0.800380     57.676887                0.761530         0.0
4      0.807977     57.650365                0.761265         0.0
5      0.805806     57.093613                0.786153         0.0
6      0.801239     56.973767                0.838859         0.0
7      0.793462     56.943421                0.796106         0.0
8      0.797214     56.883453                0.802435         0.0
9      0.798941     56.780709                0.773792         0.0
```

[8]:
```
def safe_str(v):
    if pd.isna(v):
        return "NA"
    return str(v)

df_prompt["config_signature"] = df_prompt.apply(
    lambda r: " | ".join([
        safe_str(r.get("prompt_family")),
        safe_str(r.get("owner")),
        safe_str(r.get("model_key")),
        safe_str(r.get("config_label")),
        safe_str(r.get("ruleset")),
        f"temp={safe_str(r.get('gen_temperature'))}",
        f"top_p={safe_str(r.get('gen_top_p'))}",
        f"rep={safe_str(r.get('gen_repetition_penalty'))}",
        f"max_new={safe_str(r.get('gen_max_new_tokens'))}",
        f"do_sample={safe_str(r.get('gen_do_sample'))}",
        f"ngram={safe_str(r.get('gen_no_repeat_ngram_size'))}",
    ]),
    axis=1
)

display(df_prompt[["config_signature"]].head(5))
```

	config_signature
0	few-shot | hariel | llama3 | llama3_cfg_3 | R1…
1	few-shot | nancy | llama3 | llama3_cfg_2 | R2 …
2	few-shot | hariel | llama3 | llama3_cfg_2 | R0…
3	few-shot | nancy | llama3 | llama3_cfg_2 | R1 …
4	few-shot | hariel | llama3 | llama3_cfg_1 | R2…

[9]:
```
ranking_prompting = df_prompt.sort_values(
    by=["leader_score", "sari", "bertscore_f1"],
    ascending=False
).reset_index(drop=True)

display(
    ranking_prompting[
        [c for c in [
            "prompt_family",
            "owner",
            "model_key",
            "config_label",
            "ruleset",
            "gen_temperature",
            "gen_top_p",
            "gen_repetition_penalty",
            "gen_max_new_tokens",
```

```
            "gen_do_sample",
            "gen_no_repeat_ngram_size",
            "sari",
            "bertscore_f1",
            "rougeL_f",
            "compression_ratio_eval",
            "exact_copy",
            "leader_score",
        ] if c in ranking_prompting.columns]
    ].head(25)
)

ranking_prompting.to_csv(OUT_DIR / "ranking_prompting_global.csv", index=False,␣
 ↪encoding="utf-8-sig")
print(OUT_DIR / "ranking_prompting_global.csv")
```

```
  prompt_family   owner model_key          config_label ruleset  \
0      few-shot  hariel    llama3          llama3_cfg_3      R1
1      few-shot   nancy    llama3          llama3_cfg_2      R2
2      few-shot  hariel    llama3          llama3_cfg_2      R0
3      few-shot   nancy    llama3          llama3_cfg_2      R1
4      few-shot  hariel    llama3          llama3_cfg_1      R2
5     zero-shot  hariel    llama3   hariel_llama3_cfg_1      R0
6     zero-shot   nancy   mistral   nancy_mistral_cfg_2      R4
7      few-shot  hariel    llama3          llama3_cfg_1      R4
8      few-shot   nancy    llama3          llama3_cfg_1      R0
9      few-shot   nancy    llama3          llama3_cfg_1      R4
10     few-shot  hariel    llama3          llama3_cfg_3      R0
11     few-shot  hariel    llama3          llama3_cfg_2      R1
12    zero-shot  hariel   mistral  hariel_mistral_cfg_1      R4
13     few-shot   nancy    llama3          llama3_cfg_1      R3
14     few-shot  hariel    llama3          llama3_cfg_1      R1
15     few-shot  hariel    llama3          llama3_cfg_3      R2
16     few-shot  hariel    llama3          llama3_cfg_1      R3
17     few-shot  hariel    llama3          llama3_cfg_2      R3
18    zero-shot  hariel    llama3   hariel_llama3_cfg_1      R1
19    zero-shot  hariel   mistral  hariel_mistral_cfg_3      R1
20     few-shot  hariel    llama3          llama3_cfg_2      R2
21     few-shot  hariel    llama3          llama3_cfg_1      R0
22     few-shot   nancy    llama3          llama3_cfg_1      R1
23    zero-shot   nancy   mistral   nancy_mistral_cfg_1      R0
24    zero-shot  hariel   mistral  hariel_mistral_cfg_3      R0

    gen_temperature  gen_top_p  gen_repetition_penalty  gen_max_new_tokens  \
0               0.3       0.90                    1.15               256.0
1               0.7       0.90                    1.10               512.0
2               0.2       0.90                    1.10               256.0
3               0.7       0.90                    1.10               512.0
```

4	0.2	0.85	1.05	256.0
5	0.2	0.85	1.05	256.0
6	0.3	0.90	1.15	400.0
7	0.2	0.85	1.05	256.0
8	0.3	0.90	1.15	256.0
9	0.3	0.90	1.15	256.0
10	0.3	0.90	1.15	256.0
11	0.2	0.90	1.10	256.0
12	0.3	0.90	1.10	256.0
13	0.3	0.90	1.15	256.0
14	0.2	0.85	1.05	256.0
15	0.3	0.90	1.15	256.0
16	0.2	0.85	1.05	256.0
17	0.2	0.90	1.10	256.0
18	0.2	0.85	1.05	256.0
19	0.2	0.85	1.10	256.0
20	0.2	0.90	1.10	256.0
21	0.2	0.85	1.05	256.0
22	0.3	0.90	1.15	256.0
23	0.7	0.90	1.10	512.0
24	0.2	0.85	1.10	256.0

	gen_do_sample	gen_no_repeat_ngram_size	sari	bertscore_f1	rougeL_f	\
0	NA	NA	42.880052	0.807136	0.417246	
1	NA	NA	42.969879	0.805323	0.398782	
2	NA	NA	43.080279	0.803035	0.411665	
3	NA	NA	42.769451	0.800380	0.418976	
4	NA	NA	42.218842	0.807977	0.443558	
5	NA	NA	41.984327	0.802757	0.405020	
6	True	4.0	41.904275	0.801483	0.396466	
7	NA	NA	41.435601	0.805806	0.421004	
8	NA	NA	41.540371	0.801239	0.377289	
9	NA	NA	42.008218	0.793462	0.363877	
10	NA	NA	41.658159	0.797214	0.378977	
11	NA	NA	41.371788	0.798941	0.391600	
12	NA	NA	41.313539	0.799005	0.383889	
13	NA	NA	41.698259	0.790053	0.360349	
14	NA	NA	40.858116	0.802526	0.405819	
15	NA	NA	41.386746	0.794378	0.368205	
16	NA	NA	40.885400	0.800278	0.403528	
17	NA	NA	41.114241	0.796844	0.380980	
18	NA	NA	40.927673	0.797565	0.394901	
19	NA	NA	40.852745	0.798477	0.413970	
20	NA	NA	41.219090	0.792370	0.382569	
21	NA	NA	40.141862	0.808021	0.422793	
22	NA	NA	40.908837	0.794884	0.372331	
23	True	4.0	40.336949	0.802013	0.408220	
24	NA	NA	40.552833	0.797208	0.408561	

	compression_ratio_eval	exact_copy	leader_score
0	0.861692	0.0	58.013464
1	0.781710	0.0	57.994847
2	0.807718	0.0	57.969580
3	0.761530	0.0	57.676887
4	0.761265	0.0	57.650365
5	0.746944	0.0	57.300870
6	0.902308	0.0	57.201885
7	0.786153	0.0	57.093613
8	0.838859	0.0	56.973767
9	0.796106	0.0	56.943421
10	0.802435	0.0	56.883453
11	0.773792	0.0	56.780709
12	1.010825	0.0	56.748324
13	0.814289	0.0	56.621079
14	0.763425	0.0	56.615891
15	0.802216	0.0	56.607148
16	0.746218	0.0	56.542357
17	0.749785	0.0	56.542317
18	0.889054	0.0	56.459191
19	1.047526	0.0	56.450713
20	0.791560	0.0	56.426270
21	0.765234	0.0	56.405961
22	0.800528	0.0	56.340652
23	0.972621	0.0	56.282678
24	0.972869	0.0	56.220031

/home/harielpadillasanchez/Documentos/TT/TT2/outputs/comparison_prompting_final/ranking_prompting_global.csv

[10]:

```python
top2_by_prompt_family = (
    ranking_prompting
    .sort_values(by=["prompt_family", "leader_score", "sari", "bertscore_f1"],␣
 ↪ascending=[True, False, False, False])
    .groupby("prompt_family", as_index=False, group_keys=False)
    .head(2)
    .reset_index(drop=True)
)

display(
    top2_by_prompt_family[
        [c for c in [
            "prompt_family",
            "owner",
            "model_key",
            "config_label",
            "ruleset",
```

```python
            "gen_temperature",
            "gen_top_p",
            "gen_repetition_penalty",
            "gen_max_new_tokens",
            "gen_do_sample",
            "gen_no_repeat_ngram_size",
            "sari",
            "bertscore_f1",
            "leader_score",
        ] if c in top2_by_prompt_family.columns]
    ]
)

top2_by_prompt_family.to_csv(OUT_DIR / "top2_by_prompt_family.csv",␣
 ↪index=False, encoding="utf-8-sig")
print(OUT_DIR / "top2_by_prompt_family.csv")
```

```
  prompt_family   owner model_key          config_label ruleset  \
0      few-shot  hariel    llama3           llama3_cfg_3      R1
1      few-shot   nancy    llama3           llama3_cfg_2      R2
2     zero-shot  hariel    llama3  hariel_llama3_cfg_1      R0
3     zero-shot   nancy   mistral  nancy_mistral_cfg_2      R4

   gen_temperature  gen_top_p  gen_repetition_penalty  gen_max_new_tokens  \
0              0.3       0.90                    1.15               256.0
1              0.7       0.90                    1.10               512.0
2              0.2       0.85                    1.05               256.0
3              0.3       0.90                    1.15               400.0

  gen_do_sample gen_no_repeat_ngram_size       sari  bertscore_f1  \
0            NA                       NA  42.880052      0.807136
1            NA                       NA  42.969879      0.805323
2            NA                       NA  41.984327      0.802757
3          True                      4.0  41.904275      0.801483

   leader_score
0     58.013464
1     57.994847
2     57.300870
3     57.201885
/home/harielpadillasanchez/Documentos/TT/TT2/outputs/comparison_prompting_final/
top2_by_prompt_family.csv
```

[11]:
```python
top3_by_prompt_and_model = (
    ranking_prompting
    .sort_values(
```

```python
        by=["prompt_family", "model_key", "leader_score", "sari",␣
↪"bertscore_f1"],
        ascending=[True, True, False, False, False]
    )
    .groupby(["prompt_family", "model_key"], as_index=False, group_keys=False)
    .head(3)
    .reset_index(drop=True)
)

display(
    top3_by_prompt_and_model[
        [c for c in [
            "prompt_family",
            "owner",
            "model_key",
            "config_label",
            "ruleset",
            "gen_temperature",
            "gen_top_p",
            "gen_repetition_penalty",
            "gen_max_new_tokens",
            "gen_do_sample",
            "gen_no_repeat_ngram_size",
            "sari",
            "bertscore_f1",
            "leader_score",
        ] if c in top3_by_prompt_and_model.columns]
    ]
)

top3_by_prompt_and_model.to_csv(OUT_DIR / "top3_by_prompt_and_model.csv",␣
↪index=False, encoding="utf-8-sig")
print(OUT_DIR / "top3_by_prompt_and_model.csv")
```

```
   prompt_family   owner model_key          config_label ruleset  \
0       few-shot  hariel    llama3          llama3_cfg_3      R1
1       few-shot   nancy    llama3          llama3_cfg_2      R2
2       few-shot  hariel    llama3          llama3_cfg_2      R0
3       few-shot  hariel   mistral         mistral_cfg_3      R0
4       few-shot  hariel   mistral         mistral_cfg_1      R4
5       few-shot  hariel   mistral         mistral_cfg_2      R3
6      zero-shot  hariel    llama3   hariel_llama3_cfg_1      R0
7      zero-shot  hariel    llama3   hariel_llama3_cfg_1      R1
8      zero-shot  hariel    llama3   hariel_llama3_cfg_3      R4
9      zero-shot   nancy   mistral   nancy_mistral_cfg_2      R4
10     zero-shot  hariel   mistral  hariel_mistral_cfg_1      R4
11     zero-shot  hariel   mistral  hariel_mistral_cfg_3      R1
```

```
    gen_temperature  gen_top_p  gen_repetition_penalty  gen_max_new_tokens  \
0               0.3       0.90                    1.15               256.0
1               0.7       0.90                    1.10               512.0
2               0.2       0.90                    1.10               256.0
3               0.2       0.85                    1.10               256.0
4               0.3       0.90                    1.10               256.0
5               0.1       0.85                    1.10               256.0
6               0.2       0.85                    1.05               256.0
7               0.2       0.85                    1.05               256.0
8               0.3       0.90                    1.15               256.0
9               0.3       0.90                    1.15               400.0
10              0.3       0.90                    1.10               256.0
11              0.2       0.85                    1.10               256.0

   gen_do_sample gen_no_repeat_ngram_size       sari  bertscore_f1  \
0             NA                       NA  42.880052      0.807136
1             NA                       NA  42.969879      0.805323
2             NA                       NA  43.080279      0.803035
3             NA                       NA  40.066895      0.799851
4             NA                       NA  40.401948      0.794802
5             NA                       NA  40.434464      0.791387
6             NA                       NA  41.984327      0.802757
7             NA                       NA  40.927673      0.797565
8             NA                       NA  40.659859      0.792474
9           True                      4.0  41.904275      0.801483
10            NA                       NA  41.313539      0.799005
11            NA                       NA  40.852745      0.798477

    leader_score
0      58.013464
1      57.994847
2      57.969580
3      56.034165
4      56.033262
5      55.916142
6      57.300870
7      56.459191
8      56.094857
9      57.201885
10     56.748324
11     56.450713

/home/harielpadillasanchez/Documentos/TT/TT2/outputs/comparison_prompting_final/
top3_by_prompt_and_model.csv
```

[12]:

```
top5_by_prompt_family = (
    ranking_prompting
```

```
    .sort_values(
        by=["prompt_family", "leader_score", "sari", "bertscore_f1"],
        ascending=[True, False, False, False]
    )
    .groupby("prompt_family", as_index=False, group_keys=False)
    .head(5)
    .reset_index(drop=True)
)

display(
    top5_by_prompt_family[
        [c for c in [
            "prompt_family",
            "owner",
            "model_key",
            "config_label",
            "ruleset",
            "sari",
            "bertscore_f1",
            "leader_score",
            "rougeL_f",
            "compression_ratio_eval",
            "exact_copy",
        ] if c in top5_by_prompt_family.columns]
    ]
)

top5_by_prompt_family.to_csv(OUT_DIR / "top5_by_prompt_family.csv",␣
↪index=False, encoding="utf-8-sig")
print(OUT_DIR / "top5_by_prompt_family.csv")
```

```
  prompt_family   owner model_key          config_label ruleset       sari  \
0      few-shot  hariel    llama3           llama3_cfg_3      R1  42.880052
1      few-shot   nancy    llama3           llama3_cfg_2      R2  42.969879
2      few-shot  hariel    llama3           llama3_cfg_2      R0  43.080279
3      few-shot   nancy    llama3           llama3_cfg_2      R1  42.769451
4      few-shot  hariel    llama3           llama3_cfg_1      R2  42.218842
5     zero-shot  hariel    llama3    hariel_llama3_cfg_1      R0  41.984327
6     zero-shot   nancy   mistral    nancy_mistral_cfg_2      R4  41.904275
7     zero-shot  hariel   mistral   hariel_mistral_cfg_1      R4  41.313539
8     zero-shot  hariel    llama3    hariel_llama3_cfg_1      R1  40.927673
9     zero-shot  hariel   mistral   hariel_mistral_cfg_3      R1  40.852745

   bertscore_f1  leader_score  rougeL_f  compression_ratio_eval  exact_copy
0      0.807136     58.013464  0.417246                0.861692         0.0
1      0.805323     57.994847  0.398782                0.781710         0.0
2      0.803035     57.969580  0.411665                0.807718         0.0
3      0.800380     57.676887  0.418976                0.761530         0.0
```

4	0.807977	57.650365	0.443558	0.761265	0.0
5	0.802757	57.300870	0.405020	0.746944	0.0
6	0.801483	57.201885	0.396466	0.902308	0.0
7	0.799005	56.748324	0.383889	1.010825	0.0
8	0.797565	56.459191	0.394901	0.889054	0.0
9	0.798477	56.450713	0.413970	1.047526	0.0

/home/harielpadillasanchez/Documentos/TT/TT2/outputs/comparison_prompting_final/top5_by_prompt_family.csv

[13]:
```
best_few = ranking_prompting[ranking_prompting["prompt_family"] == "few-shot"].
 ↪iloc[0]
best_zero = ranking_prompting[ranking_prompting["prompt_family"] ==␣
 ↪"zero-shot"].iloc[0]

print("Mejor few-shot:")
for col in [
    "prompt_family", "owner", "model_key", "config_label", "ruleset",
    "gen_temperature", "gen_top_p", "gen_repetition_penalty",␣
 ↪"gen_max_new_tokens",
    "gen_do_sample", "gen_no_repeat_ngram_size",
    "sari", "bertscore_f1", "leader_score"
]:
    if col in best_few.index:
        print(f"{col}: {best_few[col]}")

print("\n" + "-" * 50 + "\n")

print("Mejor zero-shot:")
for col in [
    "prompt_family", "owner", "model_key", "config_label", "ruleset",
    "gen_temperature", "gen_top_p", "gen_repetition_penalty",␣
 ↪"gen_max_new_tokens",
    "gen_do_sample", "gen_no_repeat_ngram_size",
    "sari", "bertscore_f1", "leader_score"
]:
    if col in best_zero.index:
        print(f"{col}: {best_zero[col]}")
```

```
Mejor few-shot:
prompt_family: few-shot
owner: hariel
model_key: llama3
config_label: llama3_cfg_3
ruleset: R1
gen_temperature: 0.3
gen_top_p: 0.9
gen_repetition_penalty: 1.15
```

```
gen_max_new_tokens: 256.0
gen_do_sample: NA
gen_no_repeat_ngram_size: NA
sari: 42.88005203397951
bertscore_f1: 0.8071358233690262
leader_score: 58.01346415514875

--------------------------------------------------

Mejor zero-shot:
prompt_family: zero-shot
owner: hariel
model_key: llama3
config_label: hariel_llama3_cfg_1
ruleset: R0
gen_temperature: 0.2
gen_top_p: 0.85
gen_repetition_penalty: 1.05
gen_max_new_tokens: 256.0
gen_do_sample: NA
gen_no_repeat_ngram_size: NA
sari: 41.98432695449868
bertscore_f1: 0.8027568548917771
leader_score: 57.30087036837028
```

[14]:

```
print("Resumen rapido prompting")
print("-" * 50)
print("Total filas combinadas:", len(df_prompt))
print("Tecnicas:", df_prompt["prompt_family"].unique().tolist())
print("Owners:", df_prompt["owner"].dropna().unique().tolist())
print("Modelos:", df_prompt["model_key"].dropna().unique().tolist())

best_global = ranking_prompting.iloc[0]

print("\nMejor config global de prompting:")
for col in [
    "prompt_family",
    "owner",
    "model_key",
    "config_label",
    "ruleset",
    "gen_temperature",
    "gen_top_p",
    "gen_repetition_penalty",
    "gen_max_new_tokens",
    "gen_do_sample",
    "gen_no_repeat_ngram_size",
```

```
    "sari",
    "bertscore_f1",
    "leader_score",
]:
    if col in best_global.index:
        print(f"{col}: {best_global[col]}")
```

```
Resumen rapido prompting
--------------------------------------------------
Total filas combinadas: 96
Tecnicas: ['few-shot', 'zero-shot']
Owners: ['hariel', 'nancy']
Modelos: ['llama3', 'mistral']

Mejor config global de prompting:
prompt_family: few-shot
owner: hariel
model_key: llama3
config_label: llama3_cfg_3
ruleset: R1
gen_temperature: 0.3
gen_top_p: 0.9
gen_repetition_penalty: 1.15
gen_max_new_tokens: 256.0
gen_do_sample: NA
gen_no_repeat_ngram_size: NA
sari: 42.88005203397951
bertscore_f1: 0.8071358233690262
leader_score: 58.01346415514875
```

[15]:
```
prompting_final_candidates = top2_by_prompt_family.copy()

display(
    prompting_final_candidates[
        [c for c in [
            "prompt_family",
            "owner",
            "model_key",
            "config_label",
            "ruleset",
            "gen_temperature",
            "gen_top_p",
            "gen_repetition_penalty",
            "gen_max_new_tokens",
            "gen_do_sample",
            "gen_no_repeat_ngram_size",
            "sari",
```

```
            "bertscore_f1",
            "leader_score",
            "rougeL_f",
            "compression_ratio_eval",
            "exact_copy",
        ] if c in prompting_final_candidates.columns]
    ]
)

prompting_final_candidates.to_csv(OUT_DIR / "prompting_final_candidates.csv",␣
 ↪index=False, encoding="utf-8-sig")
print(OUT_DIR / "prompting_final_candidates.csv")
```

```
  prompt_family   owner model_key         config_label ruleset  \
0      few-shot  hariel    llama3         llama3_cfg_3      R1
1      few-shot   nancy    llama3         llama3_cfg_2      R2
2     zero-shot  hariel    llama3  hariel_llama3_cfg_1      R0
3     zero-shot   nancy   mistral  nancy_mistral_cfg_2      R4

   gen_temperature  gen_top_p  gen_repetition_penalty  gen_max_new_tokens  \
0              0.3       0.90                    1.15               256.0
1              0.7       0.90                    1.10               512.0
2              0.2       0.85                    1.05               256.0
3              0.3       0.90                    1.15               400.0

  gen_do_sample gen_no_repeat_ngram_size       sari  bertscore_f1  \
0            NA                       NA  42.880052      0.807136
1            NA                       NA  42.969879      0.805323
2            NA                       NA  41.984327      0.802757
3          True                      4.0  41.904275      0.801483

   leader_score  rougeL_f  compression_ratio_eval  exact_copy
0     58.013464  0.417246                0.861692         0.0
1     57.994847  0.398782                0.781710         0.0
2     57.300870  0.405020                0.746944         0.0
3     57.201885  0.396466                0.902308         0.0
```

```
/home/harielpadillasanchez/Documentos/TT/TT2/outputs/comparison_prompting_final/
prompting_final_candidates.csv
```